

AAGL NBI Adenomyosis Abstract:

Study Objective: To evaluate the effectiveness of Narrow Band Imaging (NBI) during hysteroscopy in the detection of adenomyosis compared to conventional white light imaging.

Design: Single-center retrospective cohort study comparing NBI versus conventional white light imaging during hysteroscopy (2012–2025). Primary outcome was histologic evidence suggestive of adenomyosis, with secondary analyses of diagnostic performance metrics and patient subgroups. NBI findings were matched to histopathological evidence suggestive of adenomyosis.

Setting: Hysteroscopy was performed at a community-based hospital by the same MIGS surgeon in the Greater Chicagoland area.

Patients or Participants: 300 subjects were eligible for analysis based on inclusion criteria of any female >18 years old undergoing hysteroscopy. After exclusion criteria, 285 records were analyzed. Records with a prior history of endometriosis/adenomyosis (N=13) or had missing pathology records were excluded (N=2).

Interventions: NBI technology used in adjunct to conventional imaging, with targeted biopsies from areas showing suspicious findings.

Measurements and Primary Results: NBI used in 75.4% of cases (215/285), with adenomyosis confirmed in 39.3% (112/285). NBI significantly improved detection (OR: 2.47, 95% CI: 1.34–4.54, $p=0.0031$) with 84.8% sensitivity, 30.6% specificity, 75.7% NPV, and 44.2% PPV. Multivariate analysis confirmed NBI as independent predictor after controlling for age and menopausal status (aOR: 2.35, 95% CI: 1.26–4.39, $p=0.0073$). Subgroup analysis showed stronger effects in menopausal women (OR: 3.17, $p=0.0050$), especially over >60 years (OR: 13.20, 95% CI: 1.64–106.46, $p=0.0031$).

Conclusion: NBI significantly enhances adenomyosis detection during hysteroscopy, particularly in postmenopausal women >60 years. Future prospective studies are warranted to overcome inherent retrospective biases, validate this study's findings, and to evaluate the impact on clinical outcomes.

Declaration of AI and AI-Assisted Technologies in the Writing Process Statement:

During the preparation of this work the author(s) used Claude.AI in order to improve readability and grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.